

RESEARCH ARTICLE

Dynamics of the spatiotemporal correlation between Aerosol Optical Depth satellite data and PM_{2.5} surface monitoring: An exploratory approach

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ABSTRACT

PM_{2.5} is a major air pollutant with diverse natural and anthropogenic sources, which significantly impacts human health and the environment. Remote sensing serves as a complementary resource for air-monitoring networks, providing valuable information on the spatiotemporal patterns of air pollutants at a global scale. This study analyzes the feasibility of the use of MODIS MAIAC Aerosol Optical Depth (AOD) remote sensing for the estimation of surface PM_{2.5} and explores the dynamic relationship between AOD and PM_{2.5} in conjunction with regional factors such as heterogeneity of emissions sources, topography and meteorological variables in the Toluca Valley Metropolitan Area (TVMA), identified as the most polluted metropolitan area in Mexico. The five-year (2018-2023) temporal analysis during the dry seasons (Nov – May) revealed similar patterns between monthly average PM_{2.5} and AOD, with higher values observed during the Hot-Dry season. Spatially, 90% of the monitoring sites exhibited a moderate linear correlation ($0.4 < R < 0.69$) between AOD and PM_{2.5}, while one site showed a weaker correlation (0.33); associated with a heterogeneity of local emission sources and the presence of coarse-mode aerosols. Regarding meteorological factors, higher Boundary Layer Height (BLH) and lower Relative Humidity (RH), during the Hot-Dry season corresponded to stronger correlations, as increased convection facilitates the vertical mixing of particles, in contrast, higher RH during the Cold-Dry impacts aerosol optical properties. This study highlights the complex influence of emission sources and meteorological conditions on the relationship between AOD and PM_{2.5}, underscoring the importance of considering these factors when resorting to satellite data for air quality assessment.

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INTRODUCTION

Particulate matter with an aerodynamic diameter smaller than 2.5 µm, commonly known as PM_{2.5}, is a criteria air pollutant, meaning that acceptable levels of exposure and ambient air quality standards have been established [1]. According to Valle-Hernández et al. (2023), PM_{2.5} can be emitted from natural sources such as wind resuspension, volcanic activity and wildfires, and from anthropogenic sources as well: industries, vehicular emissions from combustion and road emissions; area emission sources such as homes and commercial premises with their own combustion emissions; and also, emissions from the burning of crop residue and dust resuspension from agriculture and livestock significantly contribute to PM_{2.5}. In addition to these

primary sources, PM_{2.5} is generated by secondary sources through atmospheric oxidation reactions of primary pollutants such as NO₂ and SO₂ to produce secondary aerosols, SO₄²⁻ and NO₃⁻ [2].

Given this diversity of emission sources, ambient PM_{2.5} concentrations are a major international concern. According to the World Health Organization, 99% of the global population breathes air with concentrations of pollutants that exceed the limits set in the Air Quality Guidelines (AQG) [3]. For the specific case of PM_{2.5}, these AQG limits are 15 µg/m³ as a 24-hour average and 5 µg/m³ as an annual average [4]. The limits are established on the basis of epidemiological evidence linking PM_{2.5} exposure to the exacerbation and development of allergies, respiratory, cardiovascular and neurological conditions, and even cancer [5]. The health

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effects above are related to the diverse composition of $PM_{2.5}$, which includes toxic substances such as heavy metals, Polycyclic Aromatic Hydrocarbons (PAHs), dioxins and furans, among others [6].

Fine particles also have a considerable impact on the environment. Their massive presence in the atmosphere affects the natural cycle of cloud formation: particles may serve as condensation nuclei, increasing cloud droplet concentration, but also reducing the cloud droplet effective radius (r_e), leading to smaller droplets, which reduces precipitation efficiency and prolongs cloud lifetime [7]. $PM_{2.5}$ composition can also diminish the pH of the rain, caused by the dissolution of SO_4^{2-} and NO_3^- (secondary aerosols) into the cloud droplets, forming the acids H_2SO_4 and HNO_3 , a phenomenon known as acid rain [8].

The term aerosol applies to the system composed of liquid or solid particles suspended in a gas, typically air [9]. Aerosols interact with solar radiation, scattering or absorbing it, depending on their composition. The degree to which aerosols prevent the transmission of light from the ground surface to the top of the atmosphere is known as Aerosol Optical Depth (AOD) or Aerosol Optical Thickness (AOT). AOD can also be considered a global measure of aerosols distributed within an air column and is strongly correlated with PM concentrations [10].

Aerosol remote sensing is performed using specific sensors on satellites that measure the amount of light that reaches the Earth's surface and how much of such light is reflected by aerosols [11]. These measurements are recorded as images where each pixel contains a value of the signal intensity captured by the sensor. Depending on the sensor technology and retrieval algorithm, pixel sizes can represent from 1 to 100 km. MODIS (Moderate Resolution Imaging Spectroradiometer) is a sensor installed on two NASA satellites, Terra and Aqua, which measures parameters of the Earth's land, oceans and atmosphere, including AOD. Originally developed for MODIS [12], the Multi-Angle Implementation of Atmospheric Correction (MAIAC) is an algorithm that uses time series analysis to improve atmospheric correction, cloud detection and aerosol retrieval. This retrieval algorithm combines the Aqua and Terra products and enhances their AOD measurements with the aforementioned corrections, which turns it into a high-resolution product (1 km) that has been widely used for spatiotemporal AOD analysis [13-15].

Remote sensing serves as a complementary resource for air-monitoring networks, providing valuable information on the spatiotemporal patterns of aerosols at a global scale [16]. To estimate ground-level $PM_{2.5}$, it is necessary to conduct remote calibrations by comparing MAIAC AOD values with respective $PM_{2.5}$ ground measurements from the local air-monitoring network, and also analyze the factors that influence this relationship. In this context, this study aims to evaluate the feasibility of using MODIS MAIAC AOD for $PM_{2.5}$ estimation within the Toluca Valley Metropolitan Area (TVMA). The TVMA is a critical area of interest as is the most polluted city in Mexico, with concentrations comparable to those found in Iraq, Shanghai, and Vietnam [17], and has been largely unaddressed in this specific context. While previous studies have confirmed the general relationship between satellite-derived AOD and $PM_{2.5}$, a comprehensive, multi-year analysis integrating specific regional factors—including local topography, diverse emission sources (industrial, vehicular, and agricultural), and seasonal meteorological variations—is often lacking. Consequently, this study represents a crucial first step in analyzing the feasibility of remote sensing and

explores the dynamic relationship between AOD and $PM_{2.5}$ in conjunction with the aforementioned regional factors. The findings, therefore, serve as a valuable reference case for implementing remote sensing measurements in similar complex urban environments.

MATERIALS AND METHODS

Study Area

TVMA is located in the central area of Mexico, 60 km west of Mexico City between $18^\circ 59' 07''$ and $19^\circ 34' 47''$ North latitude and $99^\circ 38' 22''$ and $99^\circ 56' 13''$ West longitude (Figure 1); it has an extension of 2,410.5 km² and has a population of 2.3 million inhabitants [18]. The poor quality of air in TVMA is attributed to several factors, including its high altitude (2610 masl) and its location within a semi-closed valley, which hinders the dispersion of air pollution [19]. Additionally, over a thousand industries are established here and there are 1.1 million registered motor vehicles in circulation [18]; for their part, agricultural fields account for over a half of the TVMA territory, and contribute with emissions from wind erosion and the burning of crop residues. The TVMA experiences three distinct climatic seasons: Cold-Dry Season (CDS) from November to February; Hot-Dry Season (HDS) from March to May; and Rainy Season (RaS) from June to October. The seasons with the highest $PM_{2.5}$ concentrations are CDS and HDS [20]. The study area was limited to the TVMA area presented in Figure 1, based on the coverage of the air-monitoring network.

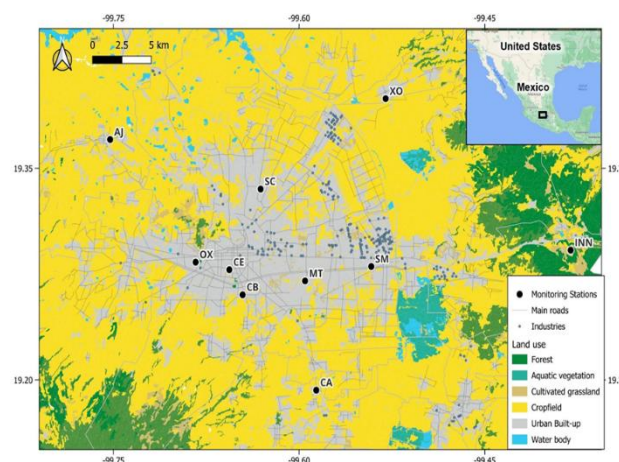


Figure 1. Air-monitoring stations, emission sources and land use in the TVMA study area

Monitoring Data

Currently, the TVMA air-monitoring network reports hourly ambient concentrations of criteria air pollutants from 8 monitoring stations. The INN monitoring station from Mexico City's air monitoring network was also included due to its proximity to TVMA. The SC site ceased operations in 2021. $PM_{2.5}$ data was limited to CDS and HDS seasons (November to May) over a five-year period (2018 to 2023). The Rainy season was avoided because of the effect of rains on AOD remote sensing such as even higher AOD missing data caused by cloud coverage and the diminishing of $PM_{2.5}$, which is partially precipitated by rain. Daily $PM_{2.5}$ concentrations were calculated as an average of the MAIAC acquisition time period, 15:00 – 18:00 (GMT-6 AM) to maintain the representativeness of the comparison between $PM_{2.5}$ and AOD, as $PM_{2.5}$ emission patterns are hourly specific (Figure 2).

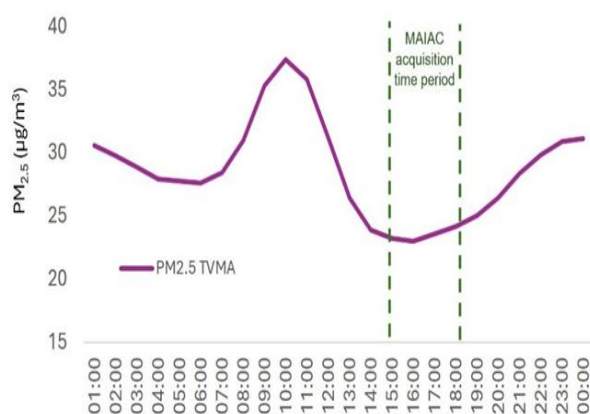


Figure 2. TVMA PM_{2.5} hourly emission profile and satellite acquisition period

Satellite Data

Land AOD spectral data was obtained from the Level 2 product MCD19A2 Version 6.1, which combines data from MODIS Terra and Aqua with the MAIAC correction algorithm, produced daily at 1 km pixel resolution [21]. The data was downloaded from the Worldview NASA browser [22], selecting the daily images of the months with the highest TVMA PM_{2.5} concentrations (Nov to May) from 2018 to 2023. The downloaded images corresponded to the granule covering the TVMA area. The Science Dataset layer used was the green band AOD at 0.55 µm. Data validation was performed on the basis of QA SDS values.

Collocation Process Between MAIAC AOD and PM_{2.5} Monitoring Data

AOD data is reported at a 1 km x 1 km grid, while ground-level PM_{2.5} monitoring data are point values. Therefore, it was necessary to collocate both variables at the monitoring stations' locations in order to assess the relationship between the surface-level PM_{2.5} and columnar AOD. The satellite images were reprojected to the WGS84 geodetic coordinate system and AOD and QA data were obtained by using the QGIS point sampling tool from the pixel corresponding to the location of each monitoring station. QA values were used to exclude pixel values with high uncertainty.

Spatiotemporal Analysis

The study employed several analyses to investigate the AOD-PM_{2.5} relationship. Spatial correlation analysis was conducted for each monitoring site (Figure 1) to identify locations with the best agreement and diagnose possible causes of low correlation. Temporal analysis evaluated changes in correlations between hot and dry seasons (November–May), it was intentionally restricted due to the severe data scarcity during the rainy season (June–October), where persistent high cloud cover and humidity resulted in AOD retrieval rates below 10%, thus violating the clear-sky assumption of the satellite algorithm. Finally, Relative Humidity (RH) and Boundary Layer Height (BLH) data were integrated into the analysis to identify and explain the meteorological factors influencing the spatiotemporal correlation between PM_{2.5} and AOD.

Meteorological Assessment

For meteorological context, Relative Humidity (RH) data were collected from the TVMA air-monitoring network at each station, while Boundary Layer Height (BLH) was retrieved from the Modern-Era Retrospective Analysis for Research and Applications (MERRA-2) product M2TMNFXFLX. Due to the coarse spatial resolution of the MERRA-2 data (0.5° × 0.625°), a direct spatial regression with the 1 km MAIAC AOD product was infeasible. Consequently, BLH was restricted to the temporal analysis, where an average TVMA value was used solely to differentiate and characterize the climatic seasons. Similarly, an average RH value was calculated for temporal characterization. However, because site-specific RH values were available, a spatial analysis was feasible using Inverse Distance Weighting (IDW) interpolation by season for comparison with the site-level AOD-PM_{2.5} correlation coefficients.

RESULTS AND DISCUSSION

MAIAC AOD Value Retrieval Rate

During the study period (2018-2023), there was a high missing rate of satellite-retrieved AOD, mostly related to orbit patterns and cloudiness [23]. The AOD retrieval percentage by site is summarized in Table 1.

As shown in Table 1, the percentage of missing AOD values was above 50% at all sites, primarily due to inconsistent orbital patterns. There was also missing data owing to cloudiness in November and December, and surface reflectivity, especially in April and May. Low AOD recovery is not unusual in remote sensing; for example, Xiao et al. (2017) also reported 40% recovery in China [24] as well as Palacios et al., (2022) documented recovery rates lower than 20% driven by the high frequency of intense cloud cover the Amazonian tropical rainforest [25]. Acknowledging the high percentage of missing values, the inherent benefits of remote sensing's continuous spatiotemporal coverage justify proceeding with the analysis. For subsequent modeling steps aimed at PM_{2.5} estimation, a robust assessment of various missing-data imputation techniques will be essential. Methods, including Chemical Transport Models (CTMs) coupled with surface measurements [26] or those detailed by Chen et al. (2020) [27], must be evaluated prior to the final modeling process. However, as an exploratory study, the dataset was utilized exactly as retrieved from the satellite product (i.e., non-imputed data). This approach establishes a clear baseline for the actual data availability and correlational factors, thereby verifying the feasibility of using satellite AOD measurements within the TVMA. AI modeling is planned as the subsequent research stage.

Temporal Variation

Similar patterns were observed between the TVMA PM_{2.5} monthly average and the AOD from MAIAC at 550 nm during the dry-seasons (Nov-May), showing a good match between both variables (Figure 3). A statistically significant, strong positive correlation was observed between the two variables using the Pearson test ($R=0.78$, $p<0.01$). The temporal variation analysis among the dry-seasons revealed that the highest PM_{2.5} and AOD values occurred in May, that is, during the HDS (March-May).

During HDS in TVMA, air pollution is exacerbated by high-pressure systems typical of this time of the year, characterized by high solar radiation and low wind speed,

which are related to stagnation of pollutants and increased formation of secondary aerosols. During this season, air quality is also diminished by agricultural burning of crop residues to prepare the land for planting just before the rainy season, as TVMA agriculture is rainfed. Forest fires are also frequent in this season, many of them caused by uncontrolled agricultural burning. CDS also presents high concentrations related to the burning of wood and charcoal for heating homes and the use of fireworks during religious festivities; all these emissions are trapped under the thermal inversions typical of this time of the year, causing pollutant accumulation. This tendency aligns well with the findings of Neria-Hernández et al. (2024) [20]. Figure 4 shows the differences between the satellite images corresponding to the CDS and HDS. The colorimetric scale of Figure 4a clearly shows higher AOD values during HDS in contrast to the CDS presented in Figure 4b, a pattern consistent with the temporal variation presented in Figure 3.

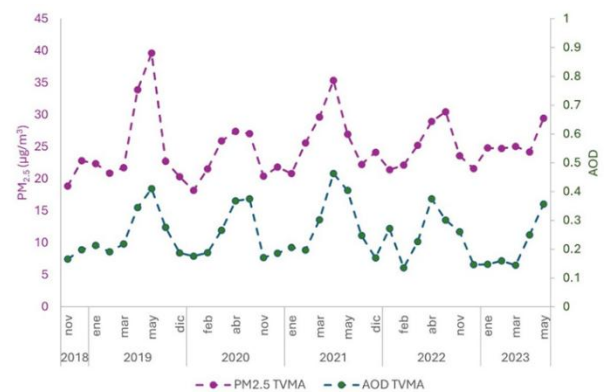


Figure 3. Monthly patterns of PM_{2.5} and MAIAC AOD (550nm) for TVMA during cold-dry season (Nov-Feb) and hot-dry season (Mar-May) from 2018 – 2023

Table 1. AOD retrieval percentage by site in TVMA during 2018-2023

Site	ININ	OX	CE	MT	CB	SM	SC	AJ	XO	CA
AOD retrieval	34%	39%	40%	41%	37%	39%	40%	42%	39%	46%

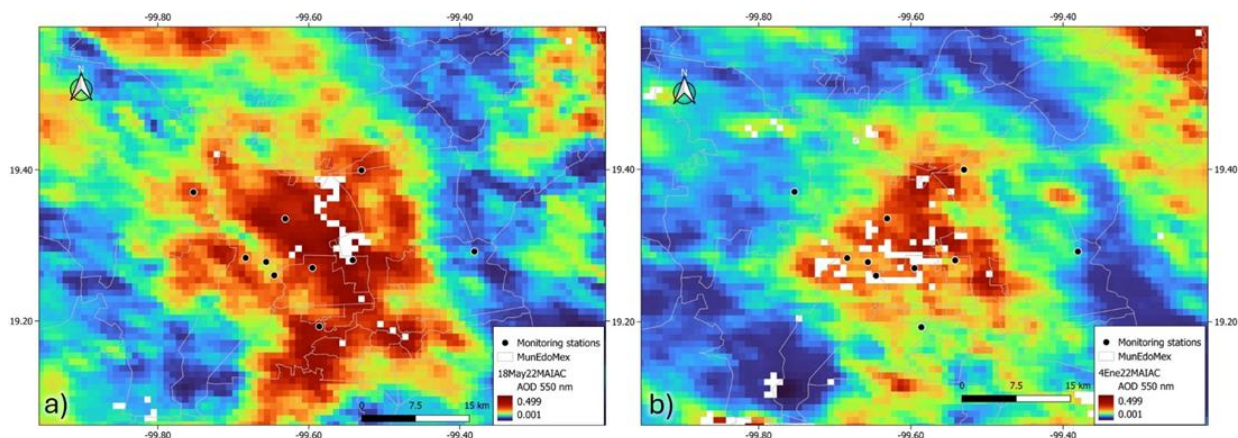


Figure 4. MAIAC AOD satellite images of TVMA during the a) Hot-Dry season (May 2nd, 2022) and b) Cold-Dry season (Jan 4th, 2022)

Correlation Between MAIAC AOD and PM_{2.5} Monitoring Data

After the collocation process of AOD data and surface PM_{2.5} at each monitoring site, a Pearson correlation assessment was conducted between both variables. The scatter plots for each site and their respective Pearson correlation values (R) are presented in Figure 5. All correlation coefficients were determined to be statistically significant using a two-tailed test ($p < 0.01$). From the 10 monitoring sites considered, 90% presented a moderate linear relationship between AOD and PM_{2.5} ($0.4 < R < 0.69$), while only one site, (SM) showed a weak correlation ($0.1 - 0.39$), this correlation coefficient interpretation was based on Schober et al., (2018) [28].

Figure 6 presents the spatial distribution of each site's Pearson correlation value, as well as land use, industrial sources and main roads in order to discuss the factors affecting the AOD and PM_{2.5} correlation values.

The sites with the highest correlations ($R > 0.6$) values (SC and MT) are located within urban and suburban areas at school districts, however, they are distant from industrial sources.

These sites are influenced by vehicular sources of aerosols and, in the case of the SC site, biomass burning from coal and wood used for cooking and agricultural burning from small parcels near the site. In contrast, the site with the weakest correlation ($R = 0.33$), SM, is affected by several sources; it is close to an industrial area (less than 3 km), main roads with heavy traffic and wind-dragged emissions from distant agricultural burnings. These facts align with the findings of Paciorek & Liu (2012) that a heterogeneity of sources at local level can contribute to weak correlations [29], as observed at the SM site.

According to Zheng et al. (2017), the presence of coarse-mode aerosols (PM₁₀) increases AOD values, diminishing their relationship with PM_{2.5} [30]. This may be the case of the CA, XO and AJ sites, located outside the urban area and surrounded by agricultural fields that remain uncovered during dry seasons, when coarse PM resuspension is frequent. The ININ site is a special case since it is located in a forest preservation area with particular climatic conditions, which will be assessed further.

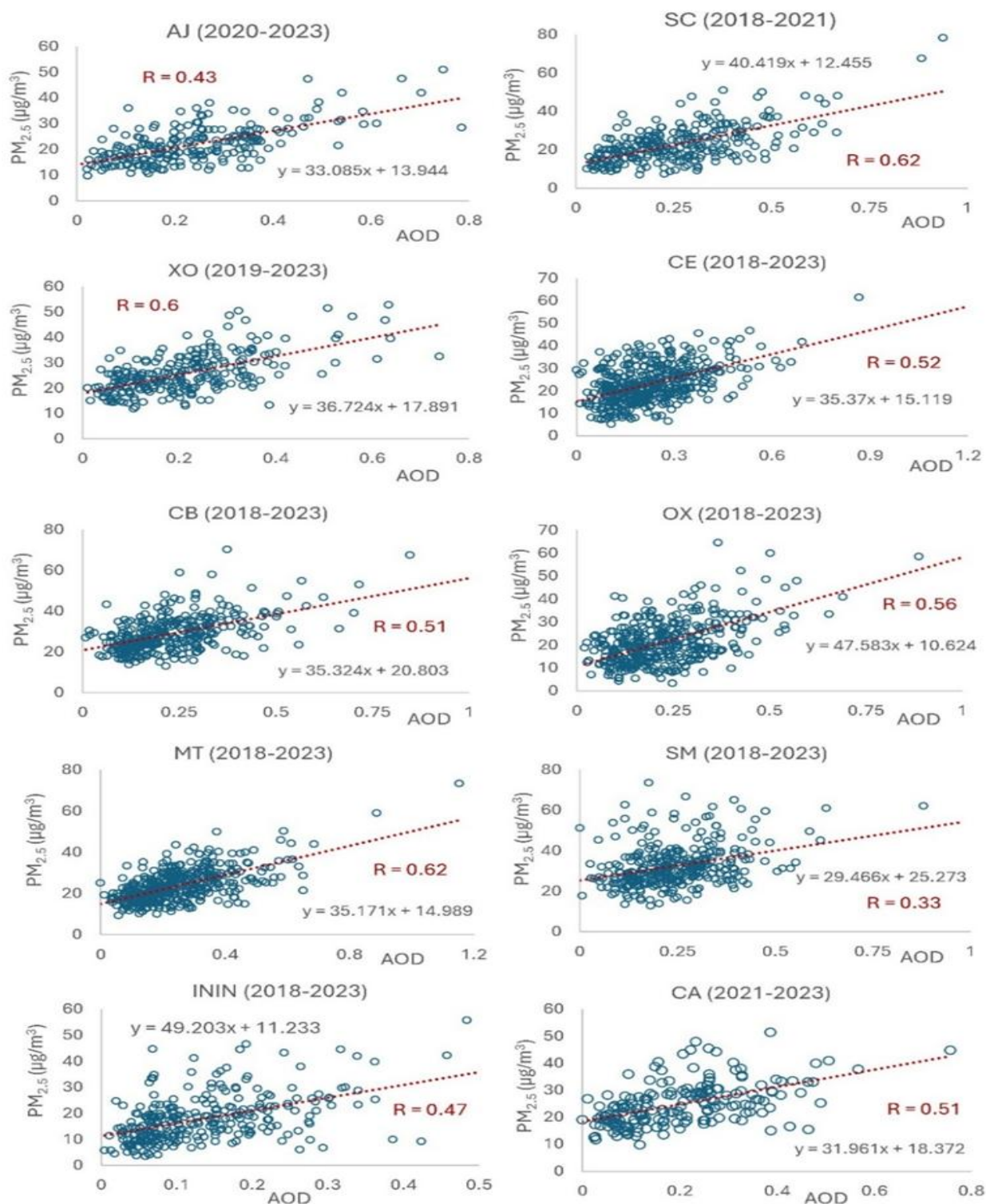


Figure 5. TVMA MAIAC AOD and surface $PM_{2.5}$ Pearson correlation ($\alpha = 0.01$) per site using daily values from Nov 2018 to May 2023

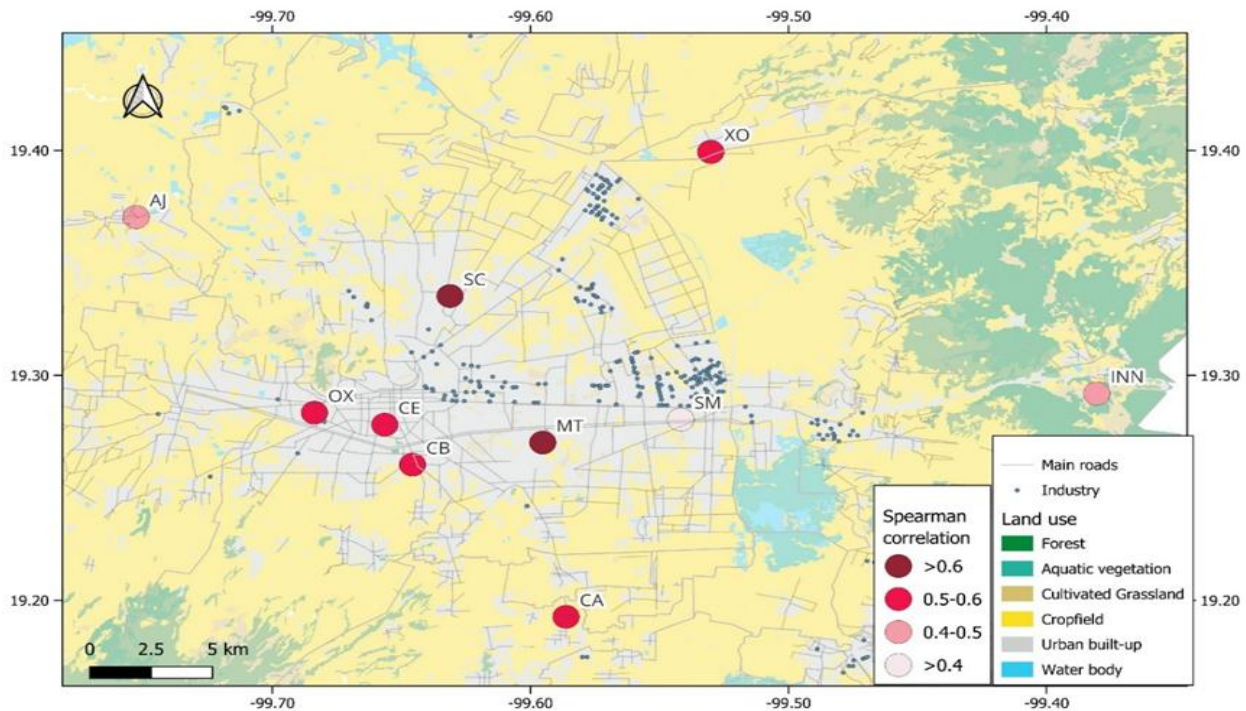


Figure 6. Spatial distribution of TVMA MAIAC AOD and surface $PM_{2.5}$ Pearson correlation values per site from Nov 2018 to May 2023

Meteorological Factors Assessment

Table 2 presents the comparison between the TVMA seasonal average values of MERRA-2 Boundary Layer Height (BLH) and surface RH with the seasonal means of the R values per site, the correlation coefficients for all seasonal analyses were found to be statistically significant ($p < 0.01$). Hot Dry Season (HDS) presented higher R values, as well as higher BLH and lower RH. In contrast, during Cold Dry Season (CDS) (November-February), RH was higher, BLH lower and correlations were lower, as shown in Table 2.

At higher BLH, there are more convection processes, which cause particles to be lifted up into the atmosphere more easily, leading to better correlations between AOD columnar measurements and $PM_{2.5}$ [31]. Conversely, in winter, at lower BLH, vertical mixing is reduced, resulting in low aerosol concentration in the upper atmosphere and high surface aerosol concentration, thus weakening the AOD and $PM_{2.5}$ correlation [32].

Regarding the effect of RH on correlation coefficients, the 2018-2023 RH average per site and season and the respective correlation values between $PM_{2.5}$ and AOD were plotted (Figure 7), resulting in a moderately negative correlation ($R = -0.54$, $p < 0.05$). RH has a high impact on the correlation between $PM_{2.5}$ and AOD, as high RH affects the water uptake process of aerosols, which can cause changes in aerosol size distribution, chemical composition and extinction characteristics [30]. As a result of the hygroscopic growth of aerosols, larger particles scatter and absorb more solar radiation (extinction) leading to a higher AOD, which includes the water absorbed by the aerosols, this can lead to an

overestimation of the actual dry aerosol concentration near the ground if not properly accounted for [33].

To analyze the spatial variability of RH, the 2018-2023 average RH was plotted for each site and interpolated using the Inverse Distance Weighting (IDW) algorithm, then compared with the R values per season, as shown in Figure 8.

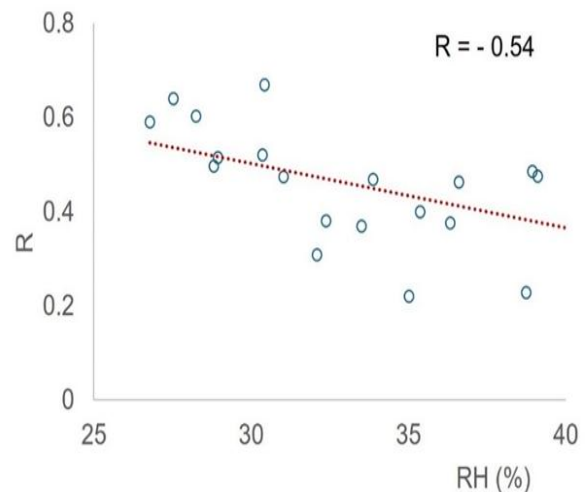


Figure 7. Scatter plot of the averages of RH vs Pearson correlation values per site and dry-season (2018-2023). ($R = -0.54$, $p < 0.05$)

Table 2. Comparison between MERRA-2 BLH and surface RH regarding the PM_{2.5} and AOD correlation by site and season

	Pearson correlation PM _{2.5} vs AOD											
	BLH (m)	RH (%)	ININ	OX	CE	MT	CB	SM	SC	AJ	XO	CA
Cold-Dry (CDS)	979.01	35.49	0.410	0.369	0.376	0.463	0.400	0.229	0.469	0.220	0.474	0.485
Hot-Dry (HDS)	1365.4	29.51	0.475	0.602	0.520	0.669	0.514	0.381	0.640	0.496	0.590	0.308
Difference (%) between seasons	28%	17%	14%	39%	28%	31%	22%	40%	27%	56%	20%	36%

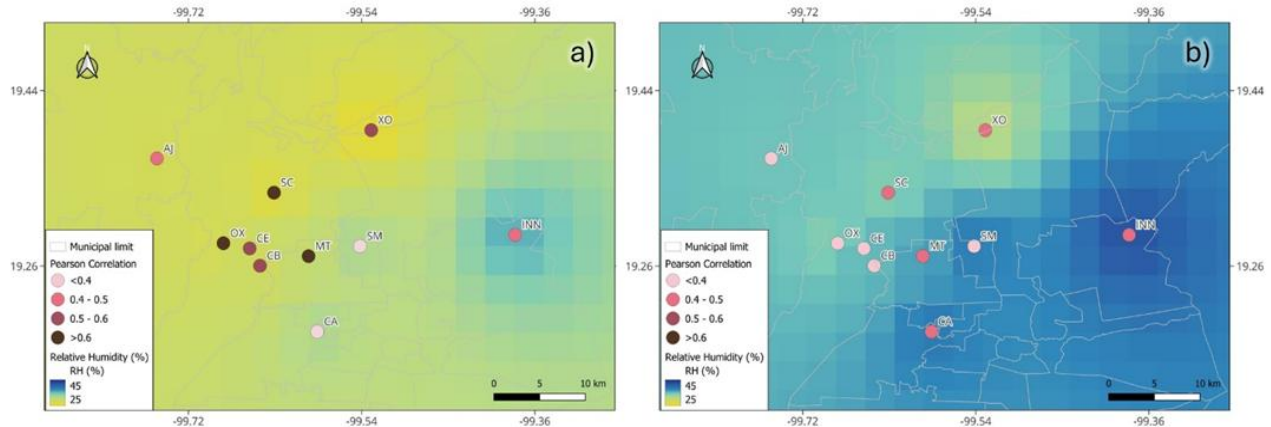


Figure 8. Average Relative Humidity per site interpolated by IDW in contrast with the Pearson correlation values ($\alpha = 0.01$) per site during a) Hot-Dry, and b) Cold-Dry seasons from 2018-2023

Regarding the relationship between the spatial variability of RH and the AOD and PM_{2.5} correlation per season (Figure 8), Pearson correlation values per site were higher during HDS than during CDS; besides, in HDS, RH was 20% lower in comparison with RH in CDS. Two sites, CA and ININ, presented atypical patterns. The CA site exhibited a higher R value in CDS than in HDS. This could be caused by the diversity of PM_{2.5} emission sources near the site in HDS, since this site is heavily impacted by agricultural burnings and wildfires and secondary aerosol formation, meanwhile during CDS, the emission is mainly from PM resuspension. In the case of the ININ site, which has the highest RH due to its location within a forest preservation area between TVMA and Mexico City, the correlation values are still moderate. This might be associated with the remote location of this site and the absence of local PM_{2.5} emission sources. Therefore, the PM_{2.5} measured at this site is likely transported by wind from urban environments. As shown in Figure 9, the ININ PM_{2.5} hourly profile differs from the TVMA profile; the five-hour difference between the maximum concentrations is evidence of the transfer of air masses with PM_{2.5}, which is well mixed, increasing the AOD and PM_{2.5} correlation. The wind rose plot, elaborated from 10:00-17:00 h, indicates that the main wind direction originates from TVMA. This has been evidenced by Hernandez-Moreno et al. (2024), who analyzed the PM_{2.5} mass exchange between three metropolitan areas in central Mexico [34].

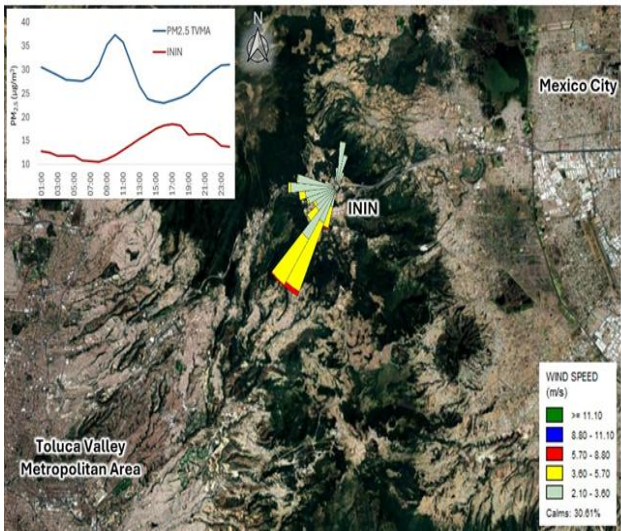


Figure 9. ININ site location, PM_{2.5} hourly profile and wind rose plot 2023

Further Steps

While our study focuses on establishing the baseline feasibility and spatiotemporal dynamics of the direct AOD–PM_{2.5} correlation, it is essential to contextualize these findings against recent methodological advancements in air quality estimation. Studies employing data fusion and machine learning (ML) models—such as Random Forest, Gradient Boosting, or Deep Learning—have significantly improved the accuracy of PM_{2.5} predictions by integrating AOD with a wide array of auxiliary inputs, including Boundary Layer Height, land-use data, and meteorological variables [26, 27, 32]. Specifically, current research demonstrates that these complex ML approaches can achieve high predictive skill

(often reporting cross-validation R^2 values exceeding 0.80) by effectively capturing the non-linear relationships ignored by simple linear correlation as reported by Lolli, (2025) [35]. Our identified moderate correlations ($R < 0.70$) across the TVMA, particularly the weaker relationships in heterogeneous industrial areas, reinforce the necessity of moving beyond simple linear models. These baseline findings serve as a critical validation step, confirming that the non-linear, locally specific factors identified here (e.g., source heterogeneity, coarse-mode aerosols) are precisely the variables that must be successfully integrated into advanced spatiotemporal ML models to achieve the high predictive accuracy required for operational air quality forecasting. The high predictive power of these applications is exemplified by Gutiérrez-Ávila et al. (2022), who developed a ML model for $PM_{2.5}$ concentration within Mexico City. The model demonstrated robust performance, achieving overall cross-validated Mean Absolute Errors (MAE) of $3.68 \mu\text{g}/\text{m}^3$ and $9.20 \mu\text{g}/\text{m}^3$ for daily mean and maximum daily 1-hour $PM_{2.5}$, respectively. This level of predictive accuracy is essential for proper health impact assessment [36].

CONCLUSIONS

This study assessed the dynamics of the spatiotemporal correlation between Aerosol Optical Depth (AOD) values obtained from MODIS MAIAC satellite data and ground-level $PM_{2.5}$ measurements, which allowed for the identification of some of the main factors that affect this relationship. Temporal analysis during the dry seasons (Nov – May) revealed a consistent monthly pattern between $PM_{2.5}$ and AOD, with both variables exhibiting higher concentrations during the Hot-Dry season (March-May). This trend is driven by meteorological conditions characteristic of this period, namely low RH and higher Boundary Layer Height (BLH).

The spatial correlation analysis revealed a moderate linear relationship between AOD and $PM_{2.5}$ ($0.4 < R < 0.69$) across 90% of the monitoring sites, with statistical significance confirmed at the 1% level ($p < 0.01$). Specifically, sites that yielded higher correlation coefficients (R ranging from 0.50 to 0.62) were generally situated in urban and suburban sectors dominated by vehicular emissions. Conversely, those locations with weaker correlations ($R = 0.33$ – 0.49) were often attributed to a more complex and heterogeneous mix of emission sources, such as contributions from industrial activity, heavy-duty road traffic, agricultural burning, and a notable presence of coarse-mode aerosols. This emphasizes that the local emission context is a crucial factor that influences the AOD- $PM_{2.5}$ relationship.

Meteorological factors, specifically BLH and RH, were identified as key influences on the correlation between $PM_{2.5}$ and AOD. Higher BLH and lower RH during the Hot-Dry season were associated with stronger correlations, likely due to enhanced vertical mixing of pollutants. Conversely, lower BLH and higher RH during the Cold-Dry season led to weaker correlations, since increased humidity can alter aerosol properties and diminish the AOD- $PM_{2.5}$ relationship. The moderate negative correlation observed between RH and the correlation coefficients further supports this finding. The atypical patterns observed at the CA and ININ sites highlight the importance of considering localized emission characteristics and air mass transport phenomena when interpreting AOD- $PM_{2.5}$ correlations.

Overall, this study confirms the applicability of MAIAC AOD data to the further estimation of $PM_{2.5}$ concentrations in the TVMA and also, at once, highlights the complex interplay of

emission sources and meteorological variables that modulate this relationship. Future studies should focus on developing more sophisticated models that integrate these influencing factors for the purpose of improving the accuracy of satellite-based $PM_{2.5}$ estimations, particularly in regions with diverse pollution sources and varying meteorological conditions.

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DATA AVAILABILITY STATEMENT

The authors confirm that the data that supports the findings of this study are available within the article. Raw data that support the finding of this study are available from the corresponding author, upon reasonable request. Satellite data originated from MODIS MAIAC is available at <https://worldview.earthdata.nasa.gov/>. The TVMA $PM_{2.5}$ Monitoring data must be requested through <https://rama.edomex.gob.mx/contactanos>.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest with any individual, institution, or organization in the preparation, evaluation, or publication of this study.

USE OF AI FOR WRITING ASSISTANCE

Not declared.

ETHICS

There are no ethical issues with the publication of this manuscript.

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